

1. Inferential Statistics

Inferential statistics means using information from a sample to understand or estimate something about a whole population. Instead of collecting data from everyone, we study a smaller group and use its results to make conclusions about the larger group. Probability is used to make these conclusions.

2. Hypothesis Testing and Its Components

Hypothesis testing is a method used to check a claim about a population with the help of sample data. It helps us decide whether the result found in the sample gives enough evidence for a particular claim.

- Null Hypothesis (H_0): This is the starting assumption in a statistical test. It usually means that there is no change, no difference, or no connection between the variables.
- Alternative Hypothesis (H_1 or H_a): This is the opposite of the null hypothesis. It states that some change, difference, or relationship exists between the variables.

3. Confidence Interval and Critical Value

- Confidence Interval (CI): A confidence interval gives a range of values that can be used to estimate a population value. For example, a 95% confidence interval gives an estimated range for the actual population parameter. It gives more information than using only one sample value.
- Critical Value: A critical value is used as a limit for taking a decision in hypothesis testing. It divides the values into a rejection area and a non-rejection area. The critical value depends on the significance level (α).

4. P-Value

The p-value is used to understand how strong the evidence is against the null hypothesis. It tells us how likely we are to get a result like the observed result when H_0 is assumed to be true.

When the p-value is 0.05 or less, we normally reject H_0 . When the p-value is more than 0.05, we normally do not reject H_0 . Therefore, the p-value helps us in making a decision about the hypothesis.

5. Type I and Type II Errors

While performing a hypothesis test, the decision made from sample data can sometimes be wrong. There are mainly two types of errors.

- Type I Error (α): This error occurs when the null hypothesis is actually true, but we reject it. It is known as a False Positive.

- Type II Error (β): This error occurs when the null hypothesis is actually false, but we do not reject it. It is known as a False Negative.

So, Type I error means rejecting a true H_0 , while Type II error means not rejecting a false H_0 .

6. Statistical Tests

Different tests are selected according to the type of data and the purpose of the analysis.

- Z-Test: A Z-test is generally used for testing population means when the sample size is large ($n \geq 30$) and the population standard deviation or variance is known.
- T-Test: A T-test is mainly used for comparing means when the sample size is small ($n < 30$) or when the population variance is unknown. It can be used for one group or for comparing two groups.
- Chi-Square Test (χ^2): This test is mainly used with categorical data. It can be used to find whether two categorical variables are related or independent. It can also compare observed frequencies with expected frequencies.
- ANOVA (Analysis of Variance): ANOVA is used when we need to compare the means of three or more groups. It checks whether there is a meaningful difference among the group means.

7. Covariance

Covariance is used to see how two variables change in relation to each other. It mainly tells us the direction in which the two variables move.

- Positive covariance: Both variables generally move in the same direction.
- Negative covariance: The variables generally move in opposite directions.
- Covariance near zero: There is little or no linear movement between the variables.

Covariance tells the direction of a relationship, but its value is not limited to a fixed range like correlation.

8. Correlation

Correlation is a statistical measure used to find the direction and strength of a linear relationship between two variables. The correlation value always lies between -1 and +1.

- +1: Shows a perfect positive relationship.
- -1: Shows a perfect negative relationship.
- 0: Shows no linear relationship.

A positive correlation means that the variables generally move in the same direction. A negative correlation means that they generally move in opposite directions. Pearson's correlation coefficient is represented by r .